

TABLE III
List of Commands

Command	Description
AT	Check whether the serial interface and GSM modem are working
ATE0	Turn echo 'off' when there is less traffic on serial line
AT+CNMI	Display the new incoming SMS
AT+CPMS	Select SMS memory
AT+CMGF	SMS string format—how they are compressed
AT+CMGR	Read the new message from a given memory location
AT+CMGS	Send message to a given recipient
AT+CMGD	Delete message

PARTS LIST

Semiconductors:

IC1	- AT89C51 microcontroller
IC2	- MAX232 driver
IC3	- 7805, 5V regulator
BR1	- 1A bridge rectifier
T1, T2	- BC548 npn transistor
LED1, LED2	- 5mm light-emitting diode
D1	- 1N4007 rectifier diode

Resistors (all 1/4-watt, ± 5 per cent carbon):

R1	- 1-k Ω -ohm
R2, R6	- 470-ohm
R3, R4	- 10-k Ω -ohm
R5	- 100-k Ω -ohm

Capacitors:

C1	- 1000 μ F, 25V electrolytic
C2, C4-C7	- 0.1 μ F ceramic
C3	- 10 μ F, 16V electrolytic
C8, C9	- 22pF ceramic

Miscellaneous:

X1	- 220V AC primary to 12V, 250mA secondary transformer
RL1	- 12V, 1C/O relay
S1	- Tactile switch
X _{TAL}	- 11.0592MHz crystal
GSM modem	- SIM300 V7.03 modem
	- Two water-level steel sensor rods

to indicate that the water level is too low to pump the water and the motor has been switched off. In this way, the motor is protected from airlocks and burnouts due to dry running. The status of motor, water level and LED2 are shown in Table I. The GSM modem used in this project is SIM300 V7.03 (refer Fig. 3). Its key features are listed in Table II.

GSM modem SIM300 V7.03

The GSM module is a specialised type of modem which accepts a SIM card

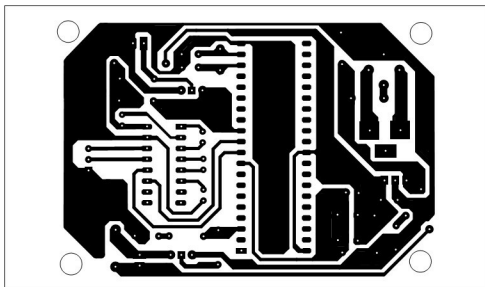


Fig. 4: An actual-size, single-side PCB for the GSM-based borewell water-level monitoring system

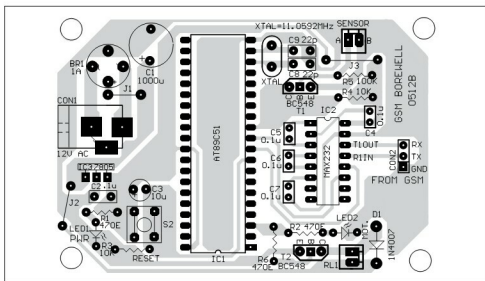


Fig. 5: Component layout for the PCB

and operates on a subscriber's mobile number over a network, just like a cellular phone. Basically, it is a cellphone without display. Modem SIM300 is a triband GSM/GPRS engine that works on EGSM 900MHz, DCS 1800MHz and PCS 1900MHz frequencies.

GSM modem is RS232-logic-level compatible, i.e., it takes -3V to -15V as logic 'high' and +3V to +15V as logic 'low'. MAX 232 is used to convert TTL into RS232 logic level and vice versa. Hence MAX232 is a voltage-level converter used between the microcontroller and the GSM board.

The signal at pin 11 of the microcontroller is sent to the GSM modem through pin 11 of MAX232. This signal is received at Pin 2 (RX) of the GSM

modem. The GSM modem transmits the signal from Pin 3 (TX) to the microcontroller through MAX232, which is received at pin 10 of IC1.

Software program

The software program is written in 'C' language and compiled using Keil software. The AT commands listed in Table III are used in the code to receive the mobile signal. The hex code of the program is burnt into the MCU using Flash Magic software.

Construction and testing

An actual-size, single-side PCB layout of the GSM-based borewell water-level monitoring system is shown in Fig. 4 and its component layout in Fig. 5.